

ES 668 Assignment 3

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25M1429

Question 1:

SO₂ is emitted in an urban area with a flux (q) of 2000 µg/m²-h. The mixing height over the area (H) is 1000 m, the atmospheric residence time (τ_r) 20 h, and SO₂ reacts with an average rate (k) of 3% h⁻¹. Rural areas around the city are characterized by a SO₂ concentration (C⁰) equal to 2 µg/m³. Assume an SO₂ dry deposition velocity (v_s) of 1 cm/s and an initial concentration (C(0)) of 2 µg/m³. What is the variation of SO₂ in the urban airshed with respect to time, according to the Eulerian and Lagrangian Box Models.

$$\text{Lagrangian Box Model: } \frac{dc_s}{dt} = \frac{q_s}{H} - kc_s - \frac{v_s}{H}c_s$$

Vary the initial concentration (C(0)) from 1 to 10 µg/m³ and check the sensitivity of the results predicted by both the models.

Answer: Script-

```
function Lagrangian_box()
clc
clear

% Initialize Parameters
q = 2000; % flux of pollutant in ug/m2-h
H = 1000; % mixing height in m
k = 0.03; % rate constant in per hour
vd = 0.01 * 3600; % m/hr
c_initial = [1:1:10]; % ug/m3, initial concentration

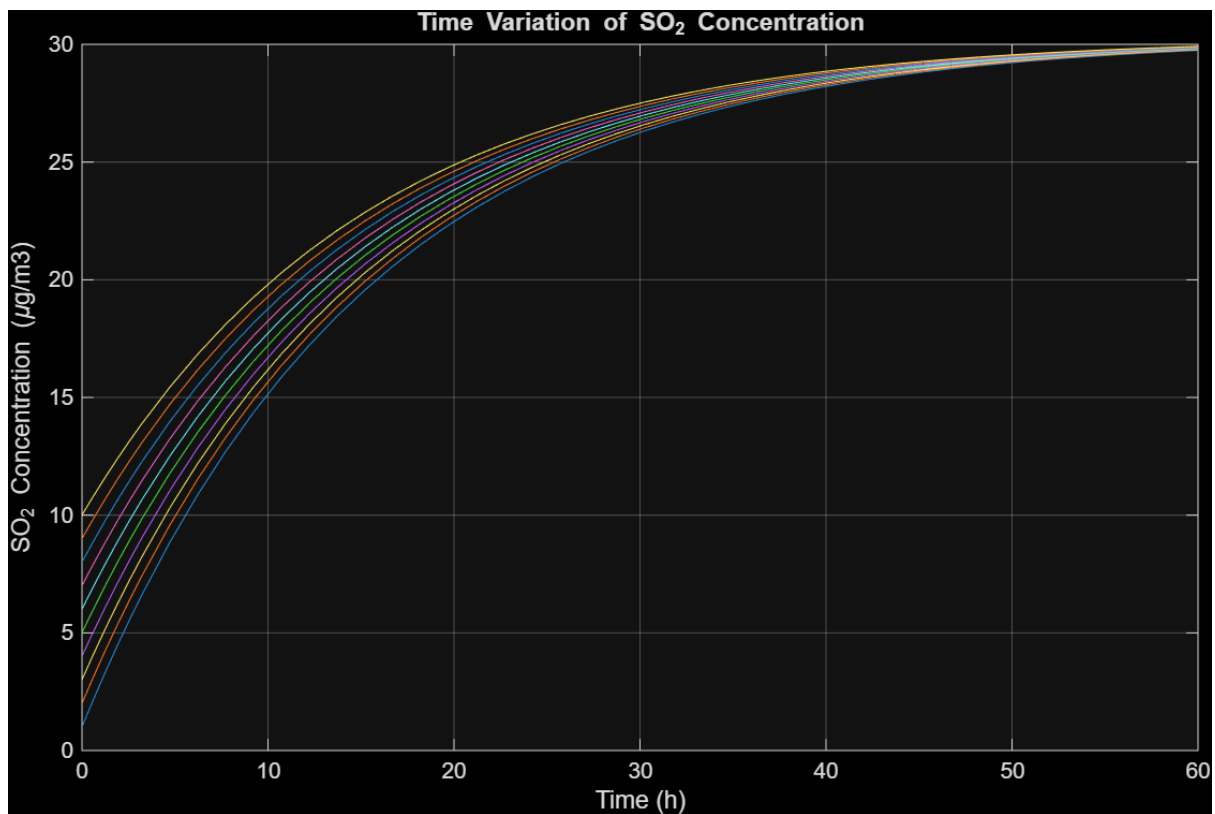
%Timespan for solving ODE
tspan = [0,60];

%Solve ODE for Lagrangian box
[t, c] = ode45(@(t,c) Lagrangian_ODE(t, c, q, H, k, vd), tspan,
c_initial);

%Plot results
plot(t, c);
xlabel('Time (h)');
ylabel('SO_2 Concentration (\mug/m3)');
title ('Time Variation of SO_2 Concentration');
grid on;
end

%Define what ODE equation is being solved in the above function
function dcdt = Lagrangian_ODE(~,c,q,H,k,vd)
dcdt = (q/H) - (k*c) - (vd*c/H)
end
```

Graph –



Question 2:

Dissolved oxygen deficit (y) in a stretch of a stream (x : length scale), due to the release of wastewater is estimated by the Streeter-Phelps ODE as follows:

$$\frac{dy}{dx} = -\left(\frac{k_a}{u}\right)y + \left(\frac{k_d}{u}\right)L_0 e^{-\left(\frac{k_d}{u}\right)x}$$

Stream velocity $u = 3$ km/day

BOD degradation rate $k_d = 1$ day⁻¹

reaeration rate $k_a = 3$ day⁻¹

initial BOD loading due to wastewater discharge $L_0 = 30$ mg/L

Initial DO deficit = 3 mg/L

Saturation DO = 8 mg/L

Using ODE45 solver in MATLAB, solve for and plot the DO deficit for a stretch of 10 km, at every 0.2 km. On the same plot, depict the actual DO levels and label the plot.

Answer : Script –

```

function Aryan_25M1429_Streeter_Phelps_box()
clear
clc

% Initialize Parameters
u = 3; % stream velocity in km/day
kd = 1; % BOD degradation rate in per day
ka = 3; % reaeration rate in per day
Lo = 30; % Initial BOD loading in mg/L
yo = 3; % Initial DO Deficit in mg/L
DO_s = 8 % Saturation DO in mg/L

% Upto 10 km stretch at every 0.2 km
xspan = [0:0.2:10];

%Solve ODE for DO Deficit
[x,y] = ode45(@(x,y) StreeterPhelps_ODE(x,y,ka,kd,u,Lo),xspan,yo);

% Do Deficit remaining
DO_act=DO_s-y;

%Plot results
plot(x,y);
hold on
plot(x,DO_act);
hold off
xlabel('Distance (in km)');
ylabel('DO Deficit Levels (mg/L)');
title ('Estimation of DO Deficit and actual DO by Streeter Phelps
ODE');
legend("DO Deficit","Actual DO",Location="east")
grid on;
end

%Define what ODE equation is being solved in the above function
function dydx = StreeterPhelps_ODE(y,x,ka,kd,u,Lo)
dydx = -(ka/u)*y + (kd/u)*Lo*exp(-(kd/u)*x)
end

```

Graph –

